

CausalLayer CALB-2 v0.8 Expansion Report

Date: 15 May 2026

Engine Version: v1.2.0

Corpus Version: v0.8.0 (250 cases)

Executive Summary

We successfully expanded the CALB-2 corpus from 202 to 250 cases, injecting 48 fresh, public, verifiable AI-incident cases. The expansion targeted previously underweight domains (aviation, automotive, government, hr_employment) and stress-tested the engine against new doctrine surfaces (defamation, copyright, and complex human-in-the-loop handovers).

Running the v1.2.0 engine blind over this expanded corpus revealed a stark but expected generalisation gap:

- **v0.7.2 baseline accuracy (n=202):** 91.6%
- **v0.8 fresh cases accuracy (n=48):** 62.5%
- **Overall v0.8 corpus accuracy (n=250):** 86.0%

This 37.5% miss rate on fresh data perfectly maps the attack surface for the v1.3 engine cycle.

Key Failure Modes for v1.3 Scoping

The 18 misses on the fresh cases cluster into several distinct, addressable failure modes:

1. The “Fake Cases” Attorney Sanction Cluster (L7-201, L7-202, L7-203, L7-204, L7-205)

The pattern: Attorneys use ChatGPT or internal LLMs to draft briefs, submit hallucinated case citations to a court, and are sanctioned under Rule 11 or state equivalents. **The engine failure:** The engine predicts `ai_provider` (Strict Product Liability) or `deployer` (Regulatory Action), but the ground truth is `human_operator` (the attorney who failed their non-delegable duty of candor to the court). **v1.3 Fix:** The FRCP-11 doctrine needs to be expanded to catch these cases and route liability 100% to the `human_operator` when the failure is `hallucination_fabrication` in a `legal_services` domain.

2. Defamation by AI / Section 230 (L1-256, L7-251)

The pattern: Pro-se plaintiffs sue AI providers (OpenAI, Microsoft) because the LLM generated defamatory statements about them. **The engine failure:** The engine predicts `ai_provider` (Strict Product Liability), but courts are routinely dismissing these under Section 230 or lack of publication, leaving the loss with the `affected_party` or `human_operator` (the prompter). **v1.3 Fix:** Implement a dedicated Publisher Liability / Defamation doctrine that routes `defamation_by_ai` cases to `human_operator` or `affected_party` depending on the jurisdiction’s Section 230 equivalent.

3. Aviation / Automotive Sensor Failures (L9-203, L9-205, L4-204)

The pattern: Complex sensor/autonomy failures (Boeing 737 MAX MCAS, Tesla Autopilot) where the NTSB/AAIB apportions blame to the manufacturer, but the engine misroutes. **The engine failure:** The engine routes to `deployer` (the airline) or `ai_provider` incorrectly because the Level-2 Automation Handover doctrine isn’t catching the specific `perception_sensor_failure` or the source types aren’t matching the aviation safety board list. **v1.3 Fix:** Broaden the Level-2 aviation branch to catch `perception_sensor_failure` and ensure NTSB/AAIB/KNKT reports trigger the correct ⁵⁰/₅₀ or provider-heavy apportionment.

4. Algorithmic Government Action (L8-201, L8-204)

The pattern: Government agencies deploy flawed algorithms (Robodebt, Allegheny Family Screening) and face regulatory or public inquiry. **The engine failure:** The engine predicts `ai_provider` (Regulatory Action), but the ground truth is `deployer` (the government agency is held responsible for the administrative failure). **v1.3 Fix:** The Government Administrative Liability doctrine needs to fire more reliably when `failure_category` is `algorithmic_government_action`, overriding the standard Regulatory Action rule.

Cryptographic Anchor

The v0.8 run has been anchored to the Bitcoin blockchain via OpenTimestamps:

- **Date:** 2026-05-15
- **Merkle Root:**
`8356b9f51ff9047e37963f9146f995d2987a6b5415c6e1bcca7a2e6278af941e`
- **Leaves:** 250
- **File:** `causallayer-anchor-log/anchors/2026-05-15-v1.2-calb2-v0.8.0.json(.ots)`

Next Steps

1. Review this report and the attached CSV of the 18 misses.
2. Approve the v1.3 scoping (specifically the FRCP-11 and Defamation doctrine additions).
3. Initiate the v1.3 engine cycle.